To Bee or not To Bee VR

Documentation Review

29th March 2019

by

An, So-Hyun
Do, Tammu
Henshaw, Benedict
Larkwright, George
Wang, Plantain

I. THE PROJECT

I.I. - WHAT IS TO BEE OR NOT TO BEE VR?

To Bee or Not To Bee is a virtual reality experience that places you in the body of a bee, delivering a new perspective on natural life and the world around us. Players must buzz their way around a garden environment, flying from flower to flower by using a unique control combination of motion controls and head-tracking.

The goal of the game is collect nectar from the flowers, pollinating them as they go, then deliver this nectar back to a hive. As the game progresses a benevolent narrator teaches the player about bees and their importance to the everyday functions of the natural world.

I.II - PROJECT AIMS

Before starting work we outlined what we wanted to achieve with this project:

Description of Aim	**How have we achieved this?**
Create an experience that will allow players to explore the world from a **different perspective**.	- By putting the player in the role of the bee we shifted the perspective to a much smaller scale. - The environment was designed as an enlarged version of a very small space – the section of garden – to enhance this sense of decreased size. - The narrator confronts the player with information that they wouldn't otherwise think about, providing another layer to this changed perspective.
Teach players more about the **importance of bees** within the natural world.	- The gameplay is based on activities that bees do in reality: collecting nectar from flowers, and taking this back to the hive (inadvertently pollinating flowers in the process). - During the game the narrator will tell the player facts about bees, their functions, and the problems that they face.

Leverage VR technology to create a unique **embodied control scheme**.	- We designed the game with VR in mind, and the design stemmed from the control scheme, rather than trying to shoehorn a VR control scheme into the game. - Players use the motion controls as wings, flapping to move towards their goals. - We used the orientation of the HMD to steer the direction of flight. - Both of these techniques further the sense that the player IS the bee.
Appeal to a wide range of players, particularly **children** and **non-gamers**.	- The game is very easy to play with simple controls and simple aim. - The graphics are cartoony and inviting. - There is a straightforward tutorial delivered through narration and on-screen prompts. - The controls map naturally to the idea of a bee flapping its wings in flight.
Deliver a **relaxing** play experience.	- During the development of the project we pivoted away from the goal of relaxation, in favour of a more frantic and humorous experience.
Weave a **humorous narrative** into the play experience.	- We used a narrator to deliver funny dialogue, and used comedic sound effects for each interaction.

I.III - DESIGN VALUES

Before beginning work on this project we decided on a trio of design values to guide development:

- **Change of Perspective**: We wanted to use virtual reality to change a player's perspective of the world, particularly exploring the concept of smallness and how the view of the world changes when you are minute creature.

- **Exploration**: The game is centred on exploration, both of the gamespace but also an exploration of attitudes towards our natural world.

- **Calm & Casual**: Whilst many successful Virtual Reality games are quite intense, we wanted to create a calming tranquil experience. It should relaxing to play, and accessible to players who might not have a gaming background.

II. VR INTERACTION & GRAPHICS

II.I. WHY VR?

Our project aims to give the player an embodied experience. We make use of the head and controller tracking to allow the player to fly as a bee using their body, as opposed to a traditional video game input, like an analogue stick or buttons. This is a key component of our experience that would be missing without VR.

The project was also made much more engaging and enjoyable to play because of VR. It is a game that is built upon the flight mechanic. The physical interaction makes the gameplay more challenging.

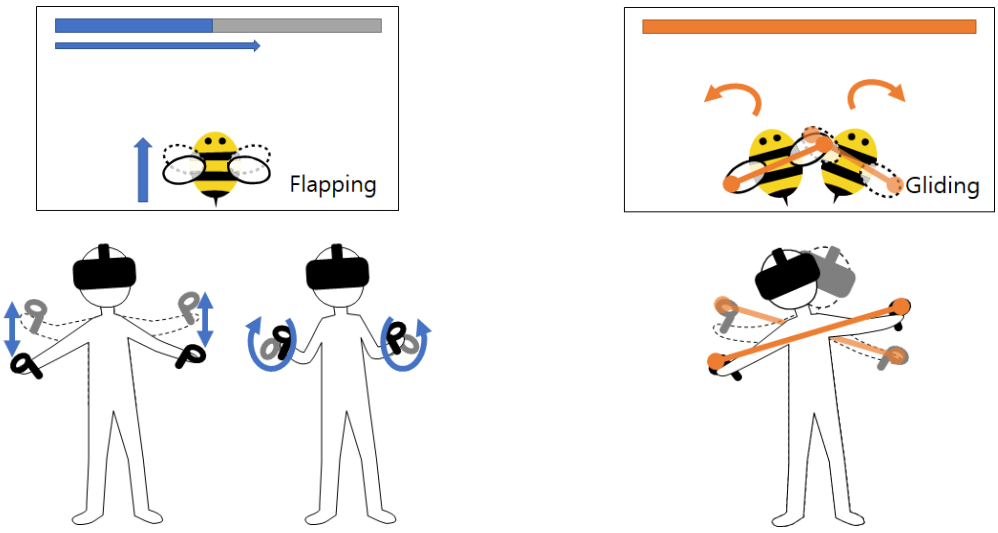
Finally the use of VR allows the player to be shrunk to the size of a bee, offering a scaled-down perspective of the world that will hopefully prompt players to think about the importance of bees.

II.II. INTERACTION EXPLORATIONS

Our game has a central mechanic — flight. We explored several different implementations of this mechanic during the course of our project. Throughout our iterations, we were guided by the goal of finding an intuitive input method, that felt simple and easy to understand, and accessible to as many people as possible.

We wanted to give users the experience of embodying the movement of bees. Bees flap their wings over 200 times a second, so we wanted to include this in our flight mechanic. However we didn't want the user to feel too tired whilst playing the game.

Our initial design for flight would ask the player to flap their hands briefly to begin flight. We considered that holding up your arms for a period of time can be tiring, so we wanted



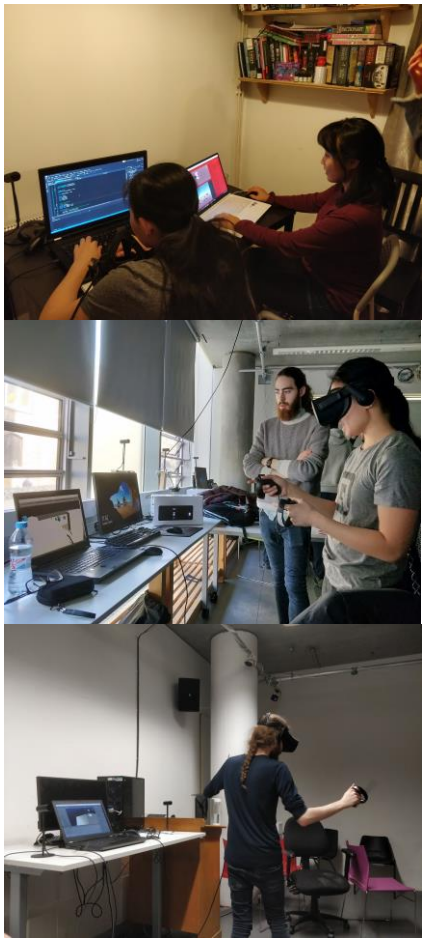
to compromise by introducing gliding after flight begins. We used a combination of their head and hands to control flight direction. This would allow six degrees of motion.

During the implementation of this design, we found that allowing the player such a high degree of control of their flight over-complicated the controls. It was hard to manage flight direction, especially when using the player's look direction — players tended to look down often and this caused them to crash into the ground. We decided to remove height control from the experience entirely, as we found that it did not affect the player's enjoyment, and allows far simpler controls.

The final iteration of the flight mechanic tracks the height of the of the controllers to detect a flapping motion, which triggers a *push* forward. This provides a simple and humorous interaction that is easy to control, with a clear and understandable outcome.

This flapping motion also contributes to how the game is perceived to non-players, as the motion is very vigorous and embodied, which is entertaining to those watching the play session.

II.III. IMPLEMENTATION



The flight mechanic is implemented by exclusively tracking the controller's Y-axis velocity. Each *flap* is one positive Y movement and one negative Y movement — moving the controller up and then down. There is an additional threshold of movement that the controller must pass in order to be detected, in order to avoid falsely detecting flaps. Once this movement has been detected the player's X and Z velocity (Y position is fixed) is updated based on the HMD's Y rotation, using straightforward trigonometry.

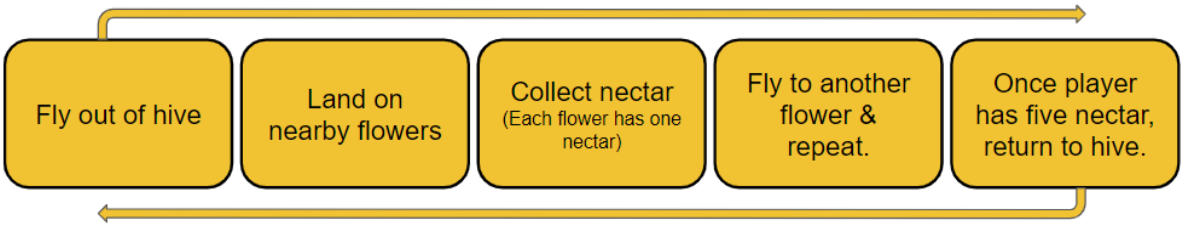
As the player's goal is to collect nectar from flowers, we used Unity's trigger volumes to detect when the player has flown into the flower, playing a sound effect, and increasing their score. This method is also used to detect when the player enters the beehive.

Simple integer counters are used to track how much nectar the player has collected — once they have visited five flowers they are instructed to drop it off at the hive, resetting the nectar counter. They must repeat this process twice to complete the game.

The basic head and hand tracking is implemented using functions from Oculus Integration for Unity. We used the OVRPlayerController example from the Unity documentation as the basis for our hand and head tracking.

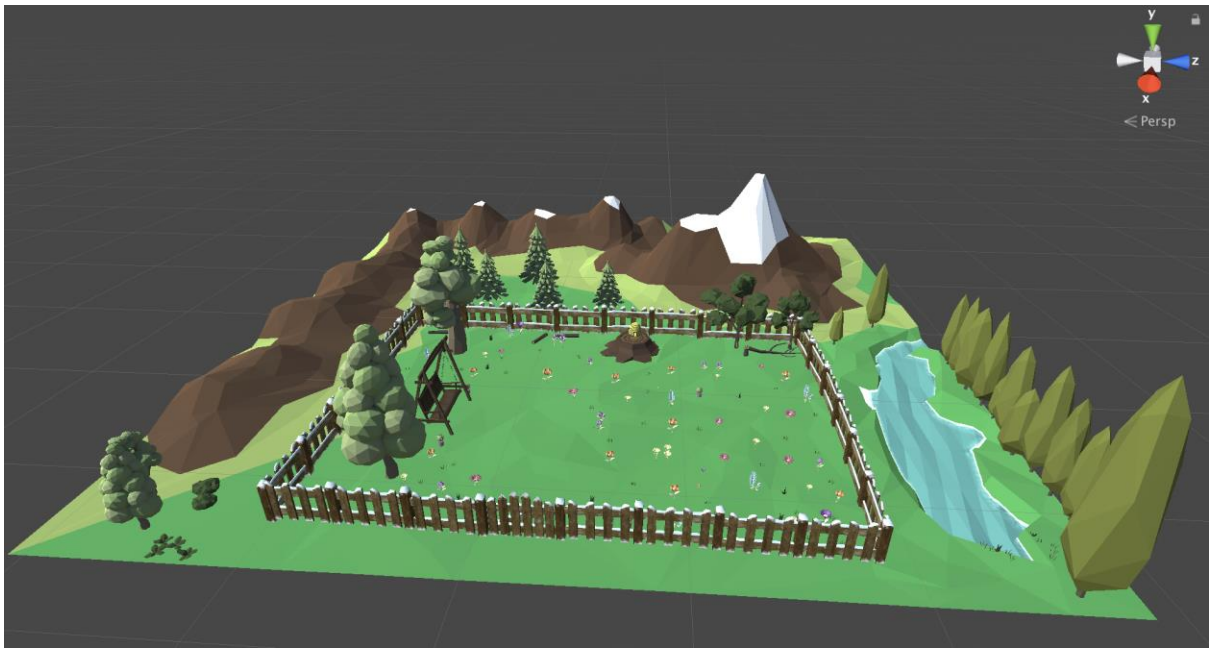
II.IV. GAMEPLAY LOOP

We decided on a simple gameplay loop to accompany the more complicated flying mechanic that we wanted to implement. This would allow us to spend more time fine-tuning the VR interactions.



1. Players start in the beehive. They must fly out of the hive to explore the world.
2. Players fly over nearby flowers to extract the nectar, adding one to their nectar score per flower.
3. Players repeat this until they have collected five nectar, at which point they will be unable to collect any more nectar and prompted to return to the hive.
4. Players return to the hive, where they drop-off their nectar before restarting the loop.

II.V. VR GRAPHICS

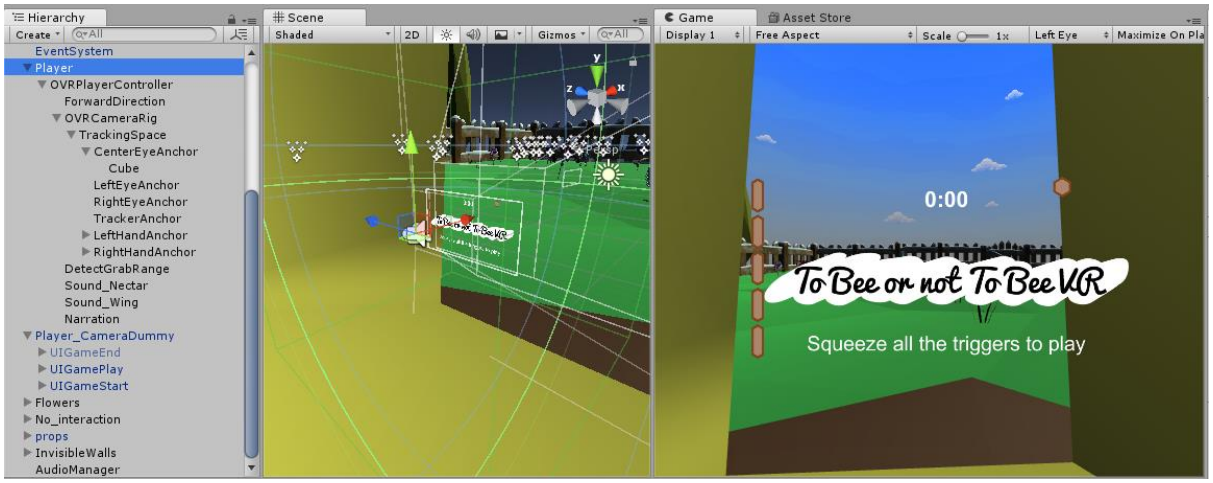


Our target audience is largely young people, so we used a low-poly cartoonish style for the game. The environment is a beautiful garden in early spring; there is some snow on the

top of mountains and fences, and a river is used to enrich the whole environment. Some trees and plants decorate the game world. Inside the garden, the hive needs to be set a little higher to help player find it easier, so it is placed on a tree stump. Because of the contrast in size with the trees or fences, the player may not feel they are really a tiny bee, so a swing is added as a comparison. Flowers are placed and we left a path leading from gate to swing.

We created a beehive as the position where the player starts the game, modelled in Maya. Other assets were taken from the Unity store.

II.VI. USER INTERFACE



User Interface in VR was totally different from on screen. The player should see all information of their inventory, goal and timer during all play time, so we designed the interface in HUD (Head Up Display) style. We could not put HUD UI on main camera like normal on-screen games, we made a dummy player which contains all UI objects and made it follow the centre camera at a proper distance.

II.VII. NARRATIVE

We decided quite early on to utilise a narrator to tell the story of our game. We wanted the narrative to reflect the motivation behind our game, to educate players on the importance of bees.

We started by drawing together as much research as bees as possible, as well as a narrative pitch.

Narrative Pitch:

You are Nigel. You don't believe in the benefits of bees. They're pests, a nuisance, and anything that destroys their habitat is a good thing.

One morning, you wake up feeling very small, and very strange. Oh lord, you've been transformed into a bee.

Guided by the voice of a sarcastic narrator, you must fly from flower to flower, pollinating each plant and discovering the significance of bees. This transformative experience will change the way you think of bees forever...

Our main player character is Nigel, a middle-aged man who hates bees. One morning he wakes up in the body of a bee - in order to change back into a human he must learn more about the importance of bees.

We weaved this narrative framework in with our research, settling on some of the key educational points we wanted the narrative to cover.

Well Nigel, let me explain…

1. The honey bee has been around for millions of years (That's longer than you, Nigel.)

2. Bees perform a vital environmental function; they are **pollinators**.

3. Pollinators transfer pollen and seeds from one flower to another, fertilising the plant so it can grow.

4. Cross-pollination is responsible for the growth of at least 30% of the world's crops.

5. Do you eat apples? Carrots? Onions? All these foods thrive thanks to cross-pollination.

6. Bees are also the only insect that produces food eaten by humans. (Unless you count spider eggs, which I don't.)

So next time you doubt the important of our little insect neighbours, think again.

These documents acted as stimuli for the writing of the game script. The script was planned to be comprised of voiceover interaction between Nigel, the player character, and the benevolent narrator whose purpose is to teach Nigel about bees. The narrator's enthusiasm combined with Nigel's angry reticence makes for great comedic potential.

We wrote two drafts of the script. The first had 43 voice lines, half of which formed a tutorial. The second draft trimmed the tutorial to 10 voice lines, and placed more emphasis on the bark lines to deliver the educational content. These lines were then recorded in Audacity before being implemented into the game. Despite the shorter script, we still managed to include plenty of interaction between Nigel and the narrator which entertained the player as they progressed through the game.

III. REFERENCES

III.I. EXISTING VR GAMES

Before starting development, we surveyed existing VR applications for inspiration. Two examples that we learned from were Be a Bee, and Eagle Flight. Both of these apps use the joysticks for flight control, which we wanted to avoid. But, they provided a good point of reference about what the experience could be like.

III.II. ASSETS AND RESOURCES

Swing model: https://assetstore.unity.com/packages/3d/props/exterior/swing-19032
Tree Stump: https://assetstore.unity.com/packages/3d/vegetation/lowpoly-trees-and-rocks-88376
Skybox: https://assetstore.unity.com/packages/vfx/shaders/free-skybox-cubemap-extended-107400
Trees and Pot: https://assetstore.unity.com/packages/3d/environments/roman-city-low-poly-pack-1-lite-92731
Trees: https://assetstore.unity.com/packages/3d/vegetation/trees/free-trees-103208
Plants: https://assetstore.unity.com/packages/3d/vegetation/low-poly-nature-free-vegetation-134006
Fences: https://assetstore.unity.com/packages/3d/environments/snowed-fence-6722
Low-poly Water: https://assetstore.unity.com/packages/tools/particles-effects/lowpoly-water-107563
Flowers: https://assetstore.unity.com/packages/3d/vegetation/plants/lowpoly-flowers-47083
Tutorial on low-poly terrain: https://www.youtube.com/playlist?list=PL7S-IAgf3dlUo25TKIjzH89LEmv2zoJgs